

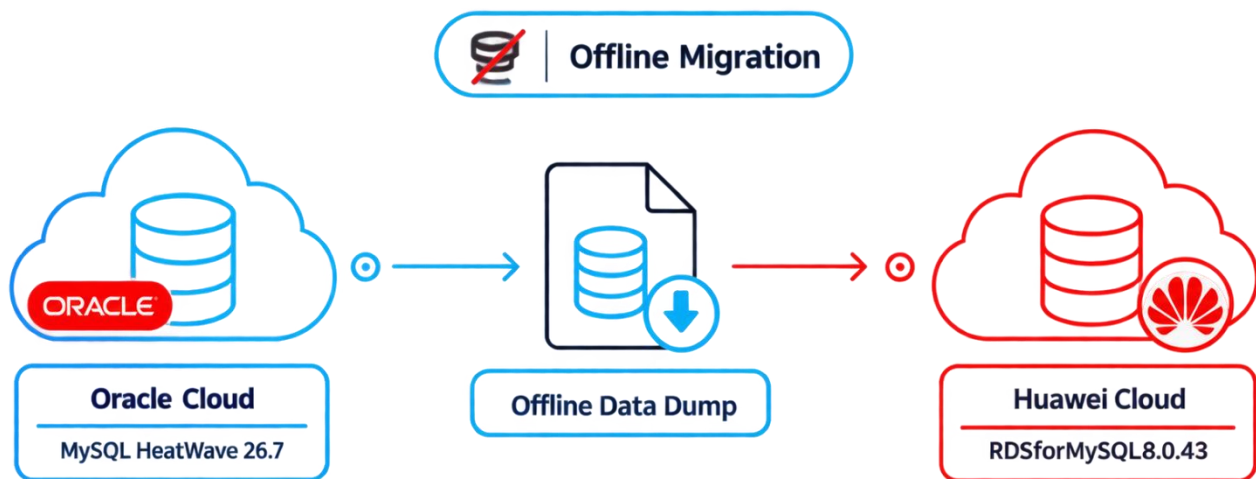


GUIDE

OFFLINE DATA DUMP MIGRATION FROM ORACLE MYSQL 26 TO RDS 8.0

An offline, step-by-step walkthrough using MySQL Shell, Huawei OBS, and Huawei RDS for MySQL

Region used in this guide: AF-Johannesburg



BY:

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Overview

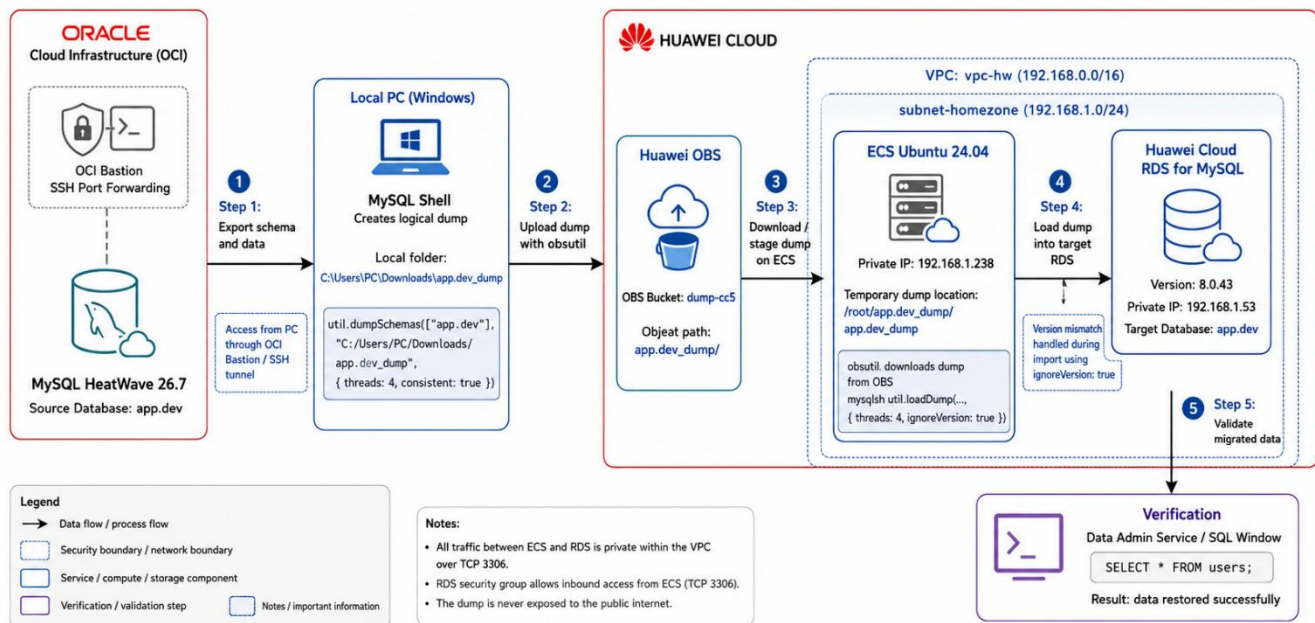
This lab shows you how to move a MySQL database out of Oracle Cloud Infrastructure (OCI), which means: Oracle's cloud computing platform, and into Huawei Cloud. The source database lives on an OCI MySQL HeatWave system, and the destination is a Huawei Cloud RDS for MySQL instance, where RDS means: Relational Database Service, a managed database that Huawei runs and maintains for you.

This guide is written for anyone doing a cloud migration for the first time, including pre-sales engineers, solution architects, and junior cloud administrators. No prior migration experience is assumed, and every tool used in this lab is introduced from the very beginning.

By the end of this lab, you will be able to export a MySQL database from OCI, carry that export safely through Huawei's storage service, and load it into a working Huawei RDS for MySQL database, all while checking your work at every stage.

Offline Cross-Cloud MySQL Migration Lab Architecture

OCI MySQL HeatWave 26.7 → Local PC → Huawei OBS → Huawei ECS → Huawei RDS for MySQL 8.0.43



NOTE: This is an offline migration, which means: the data is copied in one batch rather than being kept continuously in sync. This approach is useful when a live, continuous migration tool cannot be used.

Before you begin

Gather the following before you start this lab. Having everything ready first will save you time later.

- Access to an OCI account with a running MySQL HeatWave DB System (the source database).
- Permission to create an OCI Bastion session (this is OCI's secure doorway into private networks).
- An SSH private key file that matches your OCI Bastion setup.
- Access to a Huawei Cloud account with permission to create RDS, ECS, and OBS resources.
- A Windows machine to use as your migration workstation (this guide uses PowerShell commands).
- Administrator rights on that Windows machine, so you can install software.
- The AK and SK for your Huawei Cloud account, which means: Access Key and Secret Key, the two-part password Huawei Cloud uses to authenticate command-line tools.
- A stable internet connection, since you will be uploading and downloading a database dump file.

WARNING: Never write your AK or SK directly into a document, a chat message, or a screenshot. Treat these two values exactly like a username and password.

Step-by-step instructions

Work through the stages below in order. Each stage builds on the one before it, so do not skip ahead.

STAGE 1

Open a Secure Tunnel into OCI

Your OCI MySQL database sits on a private network, which means: it does not have a public address that you can connect to directly. OCI Bastion gives you a temporary, secure tunnel into that private network. This stage sets up that tunnel.

1. Sign in to your OCI Console.
2. Open the Bastion service.
3. Create a new Bastion session.
4. Set the session type to "SSH port forwarding session".
5. Set the target IP address to your MySQL database's private IP address, for example 10.2.0.123.
6. Set the target port to 3306, which is the standard MySQL port.
7. Create the session and wait for it to become active.

DetailsConnectionsToolsMaintenanceEndpointsRead replicasBackupsChannelsTelemetrySecurityMonitoringWork requestsTags

DB system information

Name	tobadev	Edit
OCID	...np3j46mrlgrk7kkihfdzkmbbq6eorerwdycn5kcvg2uuca	Copy
Description	—	Edit
Compartment	mapayioluwatoba (root)/migration	
Created	Sat, Aug 15, 2026, 16:41:59 UTC	
Last updated	Mon, Aug 17, 2026, 10:16:29 UTC	
Database version	26.7.0	Edit
Access mode	All users	Edit
Database mode	Read/write	Edit
Crash recovery	Enabled	Edit

HeatWave cluster

HeatWave cluster

Enabled

View details

State

Active

High availability

High availability

Disabled

Enable

What is high availability?

DB system status

Status

Not generated

Generate

Triggers an asynchronous workflow that generates runtime status summary for the DB system, aggregating information about health, connectivity, operational indicators, and transaction set on the DB system.

Work request state

None

DetailsConnectionsToolsMaintenanceEndpointsRead replicasBackupsChannelsTelemetrySecurityMonitoringWork requestsTags

Resource allocation

Shape	MySQLFree	Always Free	Edit
ECPU Count	1	Primary: 1 ECPU (1 x 1 ECPU)	
Memory	8 GiB	Primary: 8 GiB (1 x 8 GiB)	
Configuration	MySQLFree.Standalone	Always Free	Edit
Storage size	50 GiB	Primary: 50 GiB (1 x 50 GiB)	Edit
Automatic storage expansion	Disabled	Edit	

Disaster recovery

Disaster recovery protects your DB system by providing peer DB systems in a different availability domain or region.

Full stack DR

Disabled

Configure

DR protection groups

—

The list of [Disaster recovery protection groups](#) that have this DB system as a member. This list may be incomplete due to insufficient policy permissions access to a group or groups.

Encrypt

Encryption key

Oracle-managed key

Edit

Oracle manages the encryption key.

Cloud

Search resources, services, documentation, and Marketplace

South Africa Central (Johannes...

Identity & Security

Identity

Access Governance

Cloud Guard

Security Zones

Security Advisor

Threat Intelligence

Firewalls

Web Application Firewall

Certificates

Scanning

Key Management

Secret Management

Zero Trust Packet Routing

Managed Access

Bastions

Bastions let you create and manage sessions that provide authenticated users with ephemeral, timebound access to resources in the tenancy. Bastions establish secure bridge connections from preconfigured IP addresses to supported target hosts that do not have a public IP address, such as compute instances running an OpenSSH server or autonomous transaction processing databases that support SSH tunneling to an arbitrary port.

Search and Filter

Search

Applied filters

Compartment migration

Create bastion

Name	State	Bastion type	Created
Bastion202608151800	Active	Standard	Sat, Aug 15, 2026, 17:01:40 UTC

Page 1 of 1 (1 - 1 of 1 total items)

Items per page 10

Cloud

Search resources, services, documentation, and Marketplace

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Create bastion

Bastion name
Bastion202608180103

Configure networking

Target virtual cloud network compartment
migration

Target virtual cloud network
VPC

Target subnet compartment
migration

Target subnet
private

Enable FQDN Support and SOCKS5

Connect to targets on the subnet using a FQDN or IP address, or connect dynamically by SOCKS5 proxy.

CIDR block allowlist

your public ip

No matches found.

Example: 11.0.0.0/24 The IP addresses or address ranges that you want to allow to connect to target resources through SSH connections created through sessions hosted by this bastion.

Cancel

Save as stack

Create bastion

Identity & Security

Bastion

Bastion202608151800

Active

Actions

Edit

Details

Sessions

Monitoring

Security

Tags

Sessions

Search and Filter

Search

Create session

Name	Session type	Target resource	Target port	Username	State	Session TTL	Started
Session-20260817-2127	Port forwarding	10.2.0.123	3306	-	Deleted	3 hours, 00 minutes	Mon, Aug 17, 2026, 20:29:59 UTC
Session-20260817-2105	Port forwarding	10.2.0.123	3389	-	Deleted	3 hours, 00 minutes	Mon, Aug 17, 2026, 20:06:02 UTC
Session-20260815-1812	Port forwarding	10.2.0.123	3306	-	Deleted	3 hours, 00 minutes	Sat, Aug 15, 2026, 17:14:50 UTC
Session-20260815-1803	Port forwarding	10.2.0.123	22	-	Deleted	3 hours, 00 minutes	Sat, Aug 15, 2026, 17:10:36 UTC

Page 1 of 1 (1 - 4 of 4 total items)

Items per page 10

Cloud

Search resources, services, documentation, and Marketplace

South Africa Central (Johannes...

Create session

Bastion name
Bastion202608151800

Session type
SSH port forwarding session

Session name
Session-20260818-0104

Connect to the target host by using:

IP address

Instance name

IP address
10.2.1.53

Port
3306

Add SSH key

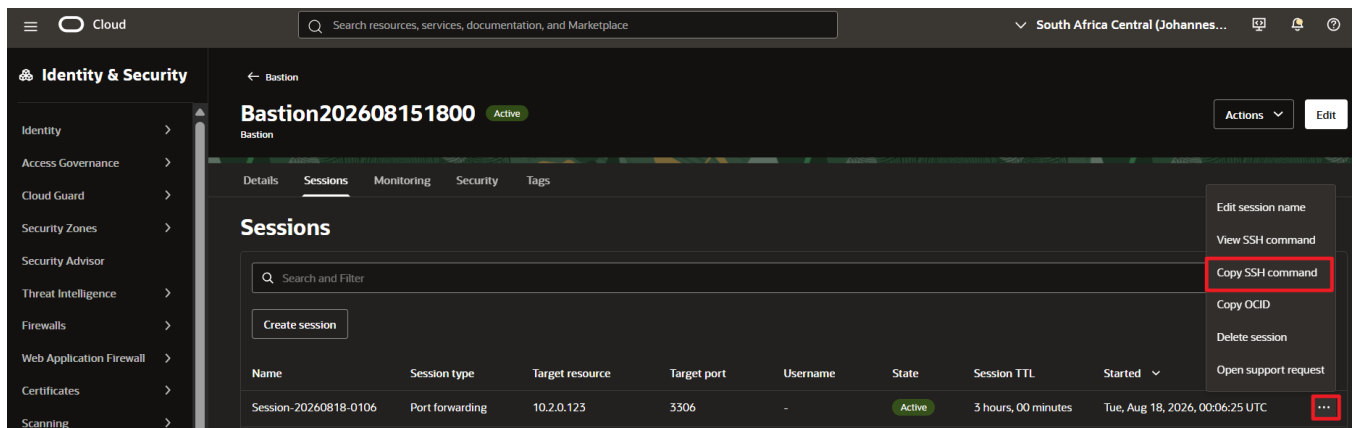
Select SSH key options:

Choose SSH key file

Paste SSH key

Cancel

Create session



If this worked, you should see: A new Bastion session listed with a status of Active.

8. Copy the SSH command that OCI provides for this session.
9. Open PowerShell on your Windows migration workstation.
10. Run the copied SSH command, replacing the key path with the location of your own private key file.

```
ssh -i "C:\path\to\private-key.key" -N `
  -L 13306:10.2.0.123:3306 `
  -p 22 `
  <BASTION_SESSION_OCID>@host.bastion.af-johannesburg-1.oci.oraclecloud.com
```

This command opens a private tunnel from port 13306 on your own computer, through OCI Bastion, and into port 3306 on the MySQL database. Port 13306 is used on your own machine instead of 3306, so that it does not clash with any MySQL software already installed on your computer.

WARNING: Leave this PowerShell window open for as long as you need to reach the OCI database. Closing it will close the tunnel.

If this worked, you should see: The PowerShell window stays open with no error message, which means the tunnel is active.

STAGE 2**Connect to the OCI Source Database**

With the tunnel open, you can now talk to the OCI database as if it were running on your own computer.

1. Open a second, new PowerShell window (keep the tunnel window from Stage 1 open).
2. Run the following command to connect with MySQL Shell, which means: a command-line tool for working with MySQL databases.

```
mysqlsh --sql -h 127.0.0.1 -P 13306 -u Admin -p
```

```

Classic protocol error: MySQL Error 2013 (HY000): Lost connection to MySQL server at 'reading initial communication pack
et', system error: 0
(base) PS C:\Users\PC> Test-NetConnection 127.0.0.1 -Port 13306

ComputerName      : 127.0.0.1
RemoteAddress     : 127.0.0.1
RemotePort        : 13306
InterfaceAlias    : Loopback Pseudo-Interface 1
SourceAddress     : 127.0.0.1
TcpTestSucceeded  : True

(base) PS C:\Users\PC> mysqlsh --sql -h 127.0.0.1 -P 13306 -u Admin -p
Please provide the password for 'Admin@127.0.0.1:13306': *****
Save password for 'Admin@127.0.0.1:13306'? [Y]es/[N]o/[e]ver (default No): Y
MySQL Shell 8.4.5

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Type '\help' or '?' for help; '\quit' to exit.
Creating a session to 'Admin@127.0.0.1:13306'
Fetching global names for auto-completion... Press ^C to stop.
Your MySQL connection id is 13328
Server version: 26.7.0-cloud MySQL Enterprise - Cloud
No default schema selected; type \use <schema> to set one.
MySQL 127.0.0.1:13306 ssl SQL >
  
```

3. Enter the password for the Admin account when prompted.
4. Once connected, run the following command to check the database version.

```
SELECT VERSION();
```

5. Run the following command to list all databases on the source server.

```
SHOW DATABASES;
```

```

Windows PowerShell x Windows PowerShell x + v
Other names may be trademarks of their respective owners.

Type '\help' or '\?' for help; '\quit' to exit.
Creating a session to 'Admin@127.0.0.1:13306'
Fetching global names for auto-completion... Press ^C to stop.
Your MySQL connection id is 13328
Server version: 26.7.0-cloud MySQL Enterprise - Cloud
No default schema selected; type \use <schema> to set one.
MySQL 127.0.0.1:13306 ssl SQL > SELECT VERSION();
+-----+
| VERSION() |
+-----+
| 26.7.0-cloud |
+-----+
1 row in set (0.2182 sec)
MySQL 127.0.0.1:13306 ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| app.dev |
| information_schema |
| mysql |
| mysql_audit |
| mysql_option |
| mysql_tasks |
| performance_schema |
| sys |
+-----+
8 rows in set (0.2106 sec)
MySQL 127.0.0.1:13306 ssl SQL >

```

If this worked, you should see: The version command returns 26.7.0-cloud, and the database list includes app.dev.

NOTE: The database name app.dev contains a period. This is unusual, and this guide will confirm later that Huawei RDS accepts this name.

STAGE 3**Install a MySQL Shell Version That Matches Your Source**

MySQL Shell needs to be the same generation as the database it is talking to, or newer. This stage checks whether your installed version is new enough, and upgrades it if it is not.

1. Attempt the database export using your current MySQL Shell version (see Stage 4).
2. Watch for the following error message.

```
Util.dumpSchemas: Unsupported MySQL Server 26.7.0 detected,
please upgrade the MySQL Shell first
```

```
8 rows in set (0.2106 sec)
MySQL 127.0.0.1:13306 ssl SQL > \js
Switching to JavaScript mode...
MySQL 127.0.0.1:13306 ssl JS > util.dumpSchemas(
    -> ["app.dev"],
    -> "C:/Users/PC/Downloads/app.dev_dump",
    -> {
    ->   threads: 4,
    ->   consistent: true
    -> }
    -> )
Util.dumpSchemas: Unsupported MySQL Server 26.7.0 detected, please upgrade the MySQL Shell first (RuntimeError)
```

WARNING: This error means your MySQL Shell tool is older than your source database. Do not try to force the export past this error; install the matching version instead.

3. Download and install MySQL Shell version 26.7 on your Windows workstation.
4. Confirm the new version by running the following command.

```
cd /root

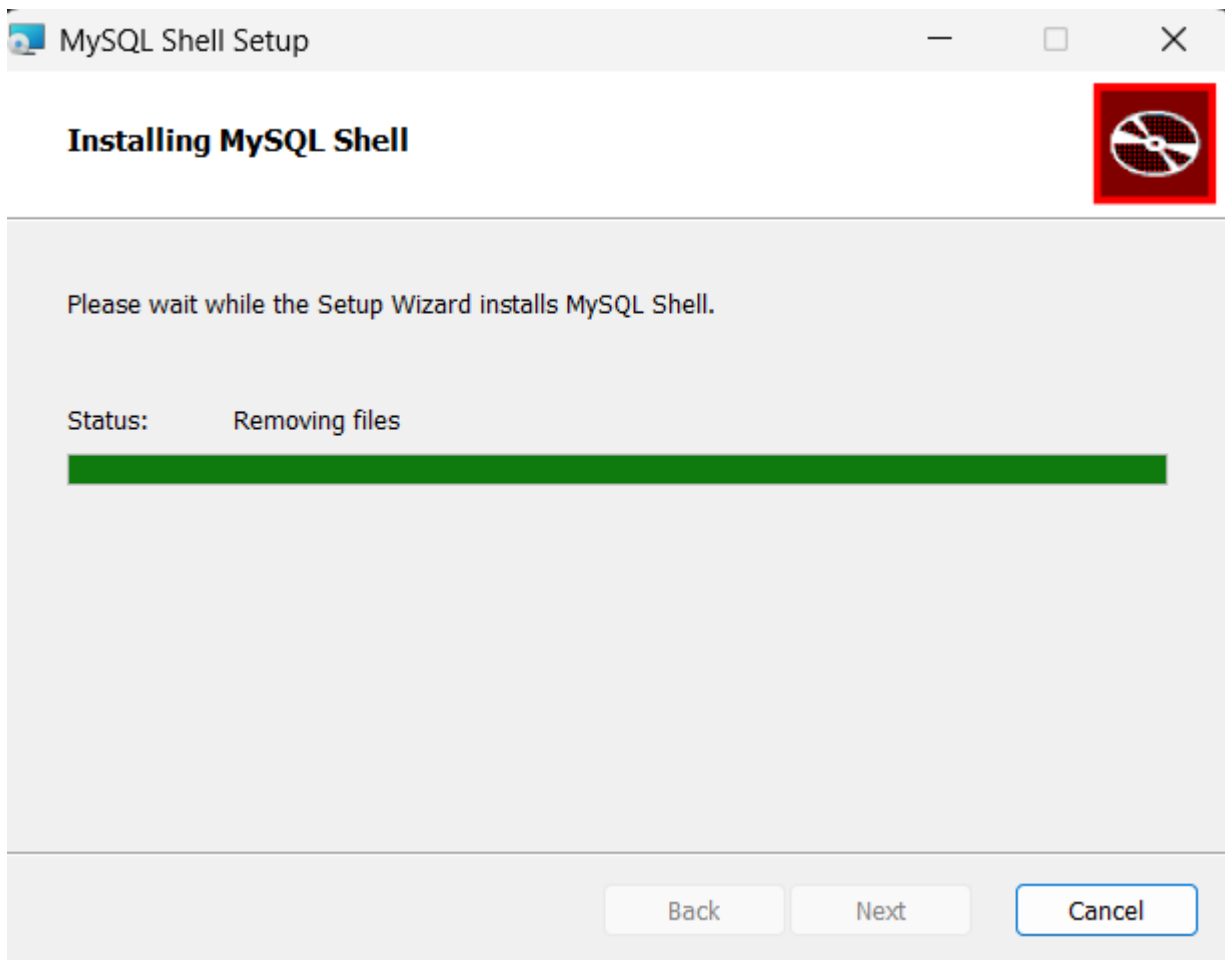
wget https://cdn.mysql.com/Downloads/MySQL-Shell/mysql-shell-26.7.0-linux-
glibc2.28-x86-64bit.tar.gz

tar -xzf mysql-shell-26.7.0-linux-glibc2.28-x86-64bit.tar.gz

mv mysql-shell-26.7.0-linux-glibc2.28-x86-64bit /opt/mysql-shell-26.7

ln -sf /opt/mysql-shell-26.7/bin/mysqlsh /usr/local/bin/mysqlsh

mysqlsh --version
```



If this worked, you should see: The version command reports MySQL Shell 26.7.0.

TIP: Always pick your MySQL Shell version to match your source database first. The destination database version matters less at this stage.

STAGE 4**Export the OCI Database to a Dump File**

This stage creates a logical dump, which means: a folder of files describing the database's structure and data, that can later be loaded into another MySQL server.

1. Reconnect to the OCI database using MySQL Shell 26.7.

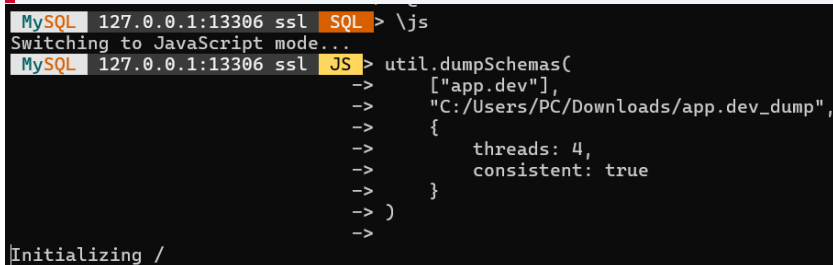
```
mysqlsh --sql -h 127.0.0.1 -P 13306 -u Admin -p
```

2. Switch MySQL Shell into JavaScript mode by typing the following.

```
\js
```

3. Run the following command to export the app.dev schema.

```
util.dumpSchemas(
  ["app.dev"],
  "C:/Users/PC/Downloads/app.dev_dump",
  {
    threads: 4,
    consistent: true
  }
)
```



```
MySQL 127.0.0.1:13306 ssl SQL > \js
Switching to JavaScript mode...
MySQL 127.0.0.1:13306 ssl JS > util.dumpSchemas(
  ->   ["app.dev"],
  ->   "C:/Users/PC/Downloads/app.dev_dump",
  ->   {
  ->     threads: 4,
  ->     consistent: true
  ->   }
  -> )
  ->
Initializing /
```

Two options in this command matter.

- threads: 4 lets the export run four parts at the same time, which makes it faster.
- consistent: true takes a single, steady snapshot of the data, so nothing changes halfway through the export.

4. Wait for the export to finish.

```

-> )
->
Acquiring global read lock
Global read lock acquired
Initializing - done
1 schemas will be dumped and within them 1 table, 0 views.
Gathering information - done
All transactions have been started
Locking instance for backup
Global read lock has been released
Writing global DDL files
Running data dump using 4 threads.
NOTE: Progress information uses estimated values and may not be accurate.
Writing schema metadata - done
Writing DDL - done
Writing table metadata - done
Starting data dump
100% (2 rows / ~2 rows), 0.00 rows/s, 0.00 B/s uncompressed, 0.00 B/s compressed
Dump duration: 00:00:00s
Total duration: 00:00:31s
Schemas dumped: 1
Tables dumped: 1
Uncompressed data size: 93 bytes
Compressed data size: 77 bytes
Compression ratio: 1.2
Rows written: 2
Bytes written: 77 bytes
Average uncompressed throughput: 93.00 B/s
Average compressed throughput: 77.00 B/s
MySQL 127.0.0.1:13306 ssl JS > |

```

If this worked, you should see: The output reports "Schemas dumped: 1" and "Rows written: 2" with no error text.

The export creates a folder containing several files, not one single file. A typical result looks like this.

```

app.dev_dump/
├─ @.json
├─ @.done.json
├─ @.sql
├─ @.post.sql
├─ app.dev.json
├─ app.dev.sql
├─ app.dev@users.json
├─ app.dev@users.sql
├─ app.dev@users@@0.tsv.zst
└─ app.dev@users@@0.tsv.zst.idx

```

NOTE: This is different from an older-style single-file export. Keep the whole folder together; every file inside it is needed for the restore later.

STAGE 5**Create the Huawei RDS for MySQL Destination**

This stage creates the destination database that will receive your data.

1. Sign in to the Huawei Cloud Console.
2. Open the RDS service.
3. Start creating a new RDS instance.
4. Set the region to AF-Johannesburg.
5. Set the database engine to MySQL.

RDS / Buy DB Instance

< Buy DB Instance ?

Basic Settings

Region ?
 AF-Johannesburg
 Regions are geographic areas isolated from each other. For low network latency and quick resource access, select the nearest region.

Billing Mode ?
 Yearly/Monthly Pay-per-use

Engine Options

DB Engine ?
 MySQL PostgreSQL MariaDB

DB Engine Version

Configuration Summary

Basic Settings
 Region | Billing Mode AF-Johannesburg | Yearly/Monthly

Engine Options
 DB Engine MySQL 8.0
 DB Instance Type Primary/Standby
 Storage Type Cloud SSD
 AZ Primary AZ: az1 | Standby AZ: az2

Instance Configuration
 Instance Specifications Dedicated | 2 vCPUs | 4 GB

6. Set the database version to MySQL 8.0, minor version 8.0.43.
7. Set the instance type to Single, which is enough for this lab.

RDS / Buy DB Instance

< Buy DB Instance ?

Instance Management

DB Instance Name ?
 rds-dump
 If you buy multiple DB instances at a time, they will be named with four digits appended in the format "DB instance name-SN". For example, if the DB instance name is "instance", the first instance will be named "instance-0001", the second "instance-0002", and so on.

Password
 Skip Configure

Administrator Administrator Password Confirm Password
 root [password] [password]

Keep your password secure. The system cannot retrieve your password. The password is checked against the instance's default password policy. The actual password policy depends on the parameter template associated with the instance. [Learn more](#)

Enterprise Project ?
 Mapayi Create Enterprise Project

Configuration Summary

Basic Settings
 Region | Billing Mode AF-Johannesburg | Yearly/Monthly

Engine Options
 DB Engine MySQL 8.0
 DB Instance Type Single
 Storage Type Cloud SSD (HomeZones)
 AZ Primary AZ: LOS1-AZ1

Instance Configuration
 Instance Specifications General-purpose

RDS / Buy DB Instance

< **Buy DB Instance** ?

Connectivity

VPC ?
vpc-hw [Create VPC](#)

The VPC an RDS instance is deployed in cannot be changed later. ECSs in different VPCs cannot communicate with each other by default.

Subnet
subnet-homezone(192.168.1.0/24) [Create Subnet](#)

⚠ An EIP is required if you want to access DB instances through a public network. [View EIP](#)

IPv4 Address
- - -

Addresses available: 249 [View IP Addresses](#)

Database Port
3306

Security Group ?

Configuration Summary

Basic Settings

Region Billing Mode	AF-Johannesburg Yearly/Monthly
-----------------------	----------------------------------

Engine Options

DB Engine	MySQL 8.0
DB Instance Type	Single
Storage Type	Cloud SSD (HomeZones)
AZ	Primary AZ: LOS1-AZ1

Instance Configuration

Instance Specifications	General-purpose
-------------------------	-----------------

8. Name the instance rds-dump, or a name of your choosing.

9. Set the database port to 3306.

10. Set a strong administrator password and store it somewhere safe.

WARNING: Never include your RDS administrator password in a screenshot, document, or chat message. Redact it before saving or sharing anything.

11. Confirm and create the RDS instance.

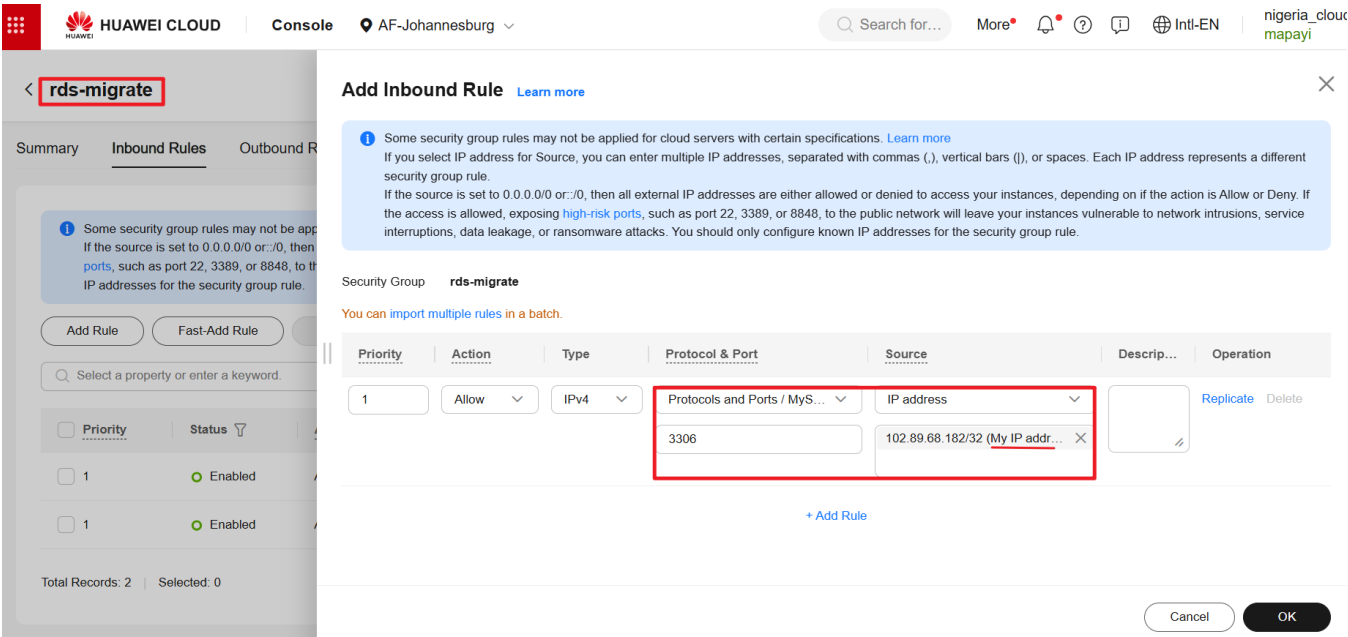
If this worked, you should see: A new RDS instance listed with a status that changes from Creating to Available after a few minutes.

STAGE 6

Configure Networking for the RDS Instance

This stage makes sure your RDS instance sits on the correct private network and can only be reached by the systems that need it.

1. Open your new RDS instance in the Huawei Console.
2. Confirm it is placed in your intended VPC, for example vpc-hw, where VPC means: Virtual Private Cloud, an isolated network you control.
3. Confirm it is placed in your intended subnet, for example subnet-homezone with address range 192.168.1.0/24.
4. Confirm the database port is set to 3306.
5. Open the security group attached to this RDS instance.
6. Add an inbound rule allowing TCP port 3306 from your migration ECS security group, or from its private IP address.



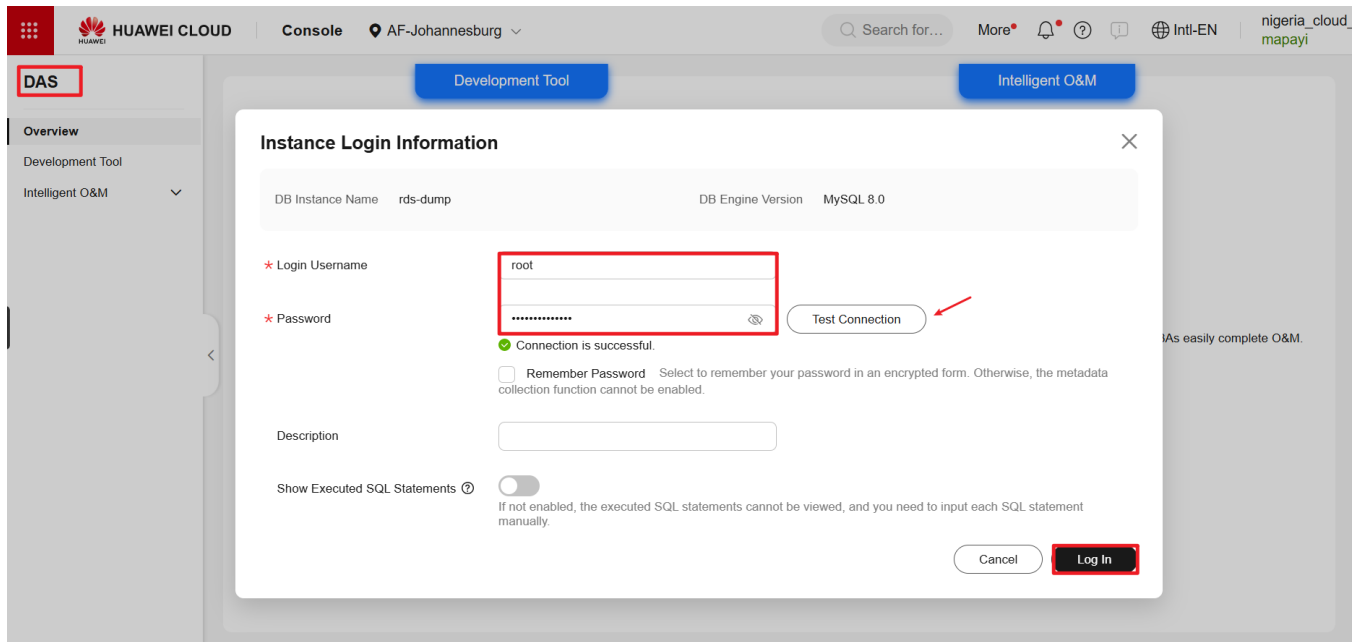
WARNING: Never open port 3306 to 0.0.0.0/0, which means: the whole internet. Only allow the specific migration ECS to reach this database.

If this worked, you should see: The security group rule list shows TCP/3306 restricted to a specific source, not to the whole internet.

STAGE 7**Check RDS Settings Using Data Admin Service**

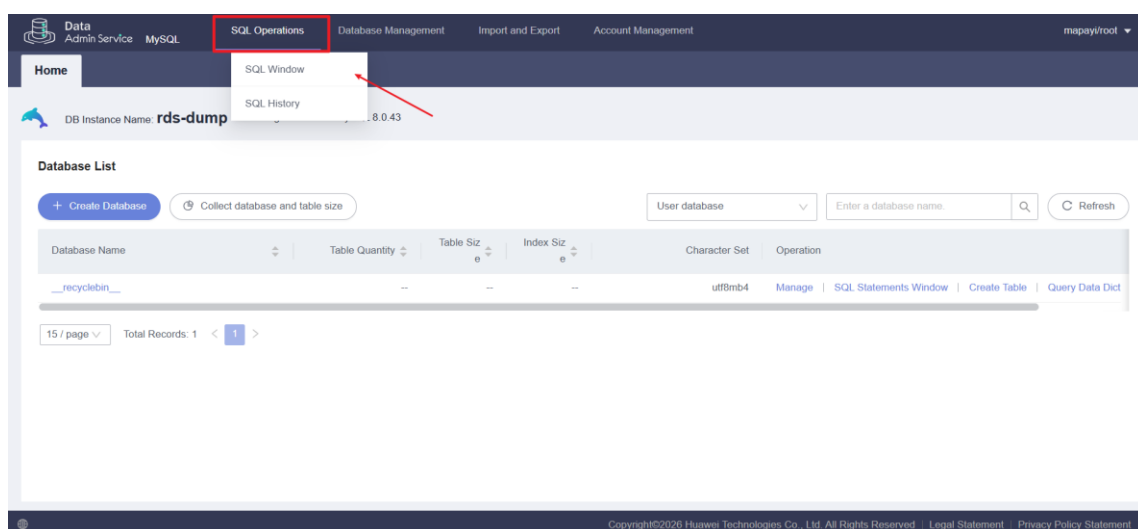
Huawei DAS, which means: Data Admin Service, lets you run SQL commands against your RDS instance from inside the Huawei Console, without installing any extra software.

1. Open the DAS service in the Huawei Console.
2. Log in to your RDS instance through DAS.



3. Run the following command.

```
SHOW GLOBAL VARIABLES LIKE 'local_infile';
```



If this worked, you should see: The result shows local_infile set to ON.

STAGE 8**Confirm the Database Name Is Accepted**

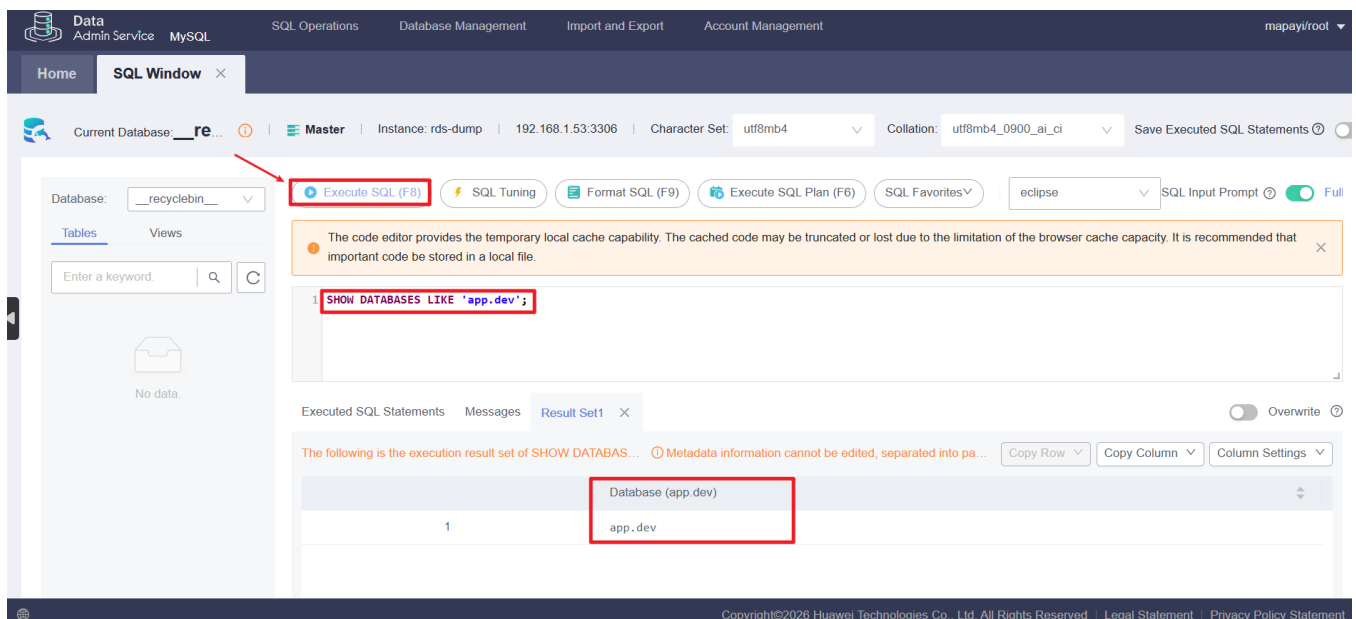
The source database name, `app.dev`, contains a period. This stage checks, before doing any real migration work, that Huawei RDS for MySQL is able to store a database with this name.

1. While still connected through DAS, run the following command.

```
CREATE DATABASE `app.dev`;
```

2. Run the following command to confirm it was created.

```
SHOW DATABASES LIKE 'app.dev';
```



If this worked, you should see: The query returns a single row containing `app.dev`.

3. Remove this test database with the following command, since it was only created to test the name.

```
DROP DATABASE `app.dev`;
```

NOTE: This test proves that the MySQL engine itself has no problem with a dotted database name. Any earlier naming problems experienced with other migration tools were related to those tools, not to MySQL on Huawei RDS.

STAGE 9**Create an OBS Bucket for the Dump File**

OBS, which means: Object Storage Service, is where your exported dump folder will be temporarily stored while it travels from your workstation into Huawei Cloud.

1. Open the OBS service in the Huawei Console.
2. Create a new bucket.
3. Name the bucket, for example dump-cc5.
4. Set the region to AF-Johannesburg, matching your RDS instance.
5. Set the storage class to Standard.

Buckets / Create Bucket

Create Bucket

General Configuration

Replicate Settings from Existing Bucket

Select Bucket

Optional. The following bucket configurations can be replicated: Region, Data Redundancy Policy, Storage Class, Block Public Access, Bucket Policy, Enterprise Project, Direct Reading, Server-Side Encryption, WORM and Tag.

Region

AF-Johannesburg

Once a bucket is created, its region **cannot** be changed. Resources are region-specific and they can be connected through an internal network within the same region. For lower latency and faster access, select the region nearest where your bucket will be accessed. [Learn how to select a region.](#)

Bucket Settings

Bucket Name

dump-cc5 [View Naming Rules](#)

Bucket domain name: dump-cc5.obs.af-south-1.myhuaweicloud.com

Data Redundancy Policy

Multi-AZ storage Single-AZ storage

This setting **cannot be changed** after the bucket is created. Data in the bucket is stored in a single availability zone (AZ). If that AZ becomes unavailable, data access may be interrupted.

6. Enable Block Public Access for the bucket and set bucket policy to private.

Buckets / Create Bucket

Create Bucket

Standard

For frequently accessed data (several times a month). It features low latency and high throughput. It supports multi-AZ (more reliable) and single-AZ (lower cost).

Cloud drives Data sharing Content sharing Hot objects

Infrequent Access

For infrequently accessed data (fewer than 12 times a year) that still needs fast retrieval. It supports multi-AZ (more reliable) and single-AZ (lower cost).

Cloud drives Enterprise backups Active archives Monitoring data

Archive

For rarely accessed data (once a year). It supports single-AZ only. To read data in this class, you must restore it or enable direct reading for the bucket.

Archives Medical imaging Video files

Block Public Access

Enabled settings: 4

Enabling Block Public Access can help prevent data leaks and theft.

Show Settings If you require public access at certain levels, you can customize the settings below to suit your specific use cases. [Learn more](#)

Bucket Policy

Private Public Read Public Read/Write Replicate Bucket Policy

The bucket owner has full control. Other users **cannot access the bucket unless authorized**. If you require public access, choose a desired bucket policy here or configure the bucket ACL after creation.

Enterprise Project

STAGE 10**Install and Configure obsutil on Windows**

obsutil is Huawei's command-line tool for moving files in and out of OBS. This stage installs it on your Windows workstation.

1. Open PowerShell on your Windows workstation.
2. Run the following commands to download and extract obsutil.

```
$dest = "$env:USERPROFILE\obsutil"

New-Item -ItemType Directory -Force -Path $dest | Out-Null

$zip = "$env:TEMP\obsutil_windows_amd64.zip"

Invoke-WebRequest `
  -Uri "https://obs-community-intl.obs.ap-southeast-
1.myhuaweicloud.com/obsutil/current/obsutil_windows_amd64.zip" `
  -OutFile $zip

Expand-Archive `
  -Path $zip `
  -DestinationPath $dest `
  -Force
```

3. Run the following command to locate the extracted obsutil.exe file and save its path for later use.

```
$obsutil = (
  Get-ChildItem $dest `
    -Recurse `
    -Filter "obsutil.exe" |
  Select-Object -First 1
).FullName
```

4. Confirm the installation by checking the version.

```
& $obsutil version
```

```

Windows PowerShell
(base) PS C:\Users\PC> # 1. Create a folder for obsutil
(base) PS C:\Users\PC> $dest = "$env:USERPROFILE\obsutil"
(base) PS C:\Users\PC> New-Item -ItemType Directory -Force -Path $dest | Out-Null
(base) PS C:\Users\PC> # 2. Download the current Windows 64-bit obsutil package
(base) PS C:\Users\PC> $zip = "$env:TEMP\obsutil_windows_amd64.zip"
(base) PS C:\Users\PC> Invoke-WebRequest `
>> -Uri "https://obs-community-intl.obs.ap-southeast-1.myhuaweicloud.com/obsutil/current/obsutil_windows_amd64.zip" `
>> -OutFile $zip
(base) PS C:\Users\PC> # 3. Extract it
(base) PS C:\Users\PC> Expand-Archive -Path $zip -DestinationPath $dest -Force
(base) PS C:\Users\PC> # 4. Find obsutil.exe
(base) PS C:\Users\PC> $obsutil = (Get-ChildItem $dest -Recurse -Filter "obsutil.exe" | Select-Object -First 1).FullName

(base) PS C:\Users\PC> # 5. Check that it works
(base) PS C:\Users\PC> & $obsutil version
obsutil version:5.8.3, obs_sdk version:3.24.12
operating system:windows_arch:amd64
(base) PS C:\Users\PC> & $obsutil config -interactive
Please input your ak:
Please input your sk:
*****
Please input your endpoint:
obs.af-south-1.myhuaweicloud.com ← From obs console
Please input your token:

```

If this worked, you should see: The command prints a version number, for example obsutil version: 5.8.3.

5. Configure obsutil to connect to the Johannesburg OBS endpoint.

```
& $obsutil config -interactive
```

6. Enter the endpoint address when prompted.

```
obs.af-south-1.myhuaweicloud.com
```

The screenshot shows the Huawei Cloud Console interface. On the left, there's a navigation menu with 'Overview', 'Basic Configurations', 'Permissions', and 'Data Security'. The main area displays the 'dump-cc5' bucket configuration. Under 'Domain Name Details', there's a table with the following data:

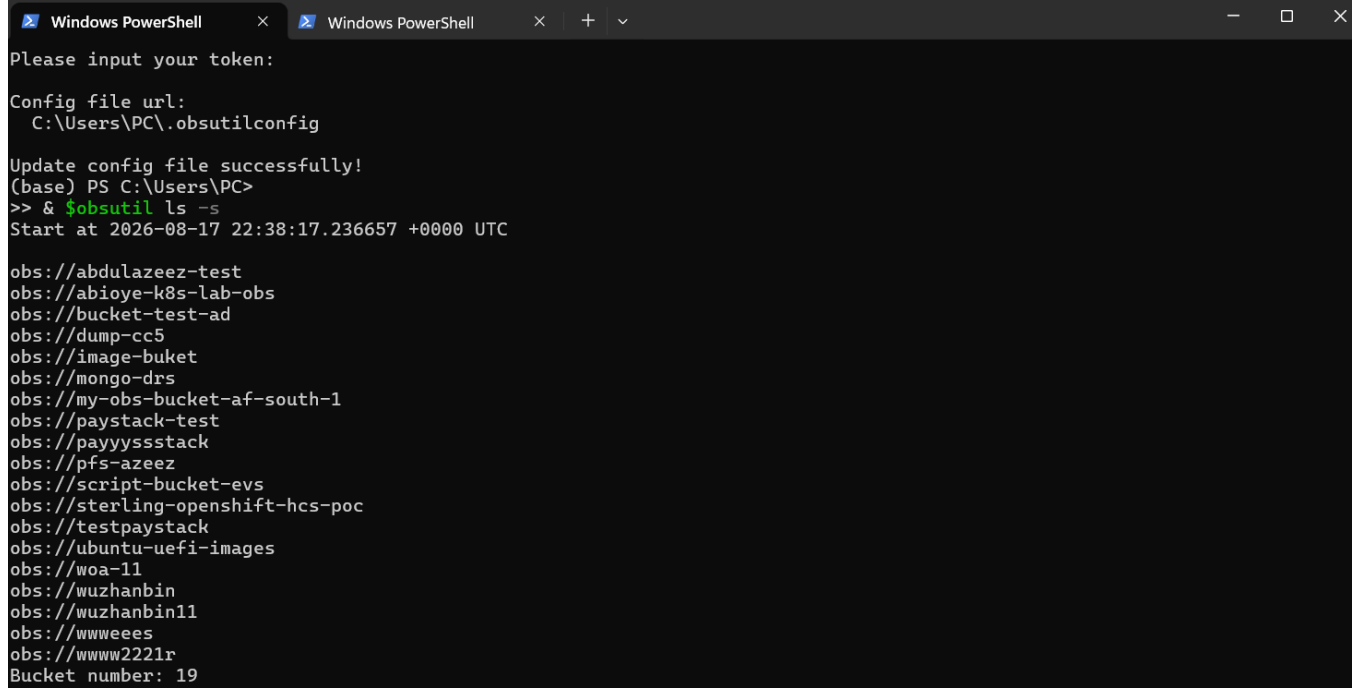
Type	Domain Name	Protocol	Operation
Endpoint	obs.af-south-1.myhuaweicloud.com	HTTPS/HTTP	--
Access Domain Name	dump-cc5.obs.af-south-1.myhuaweicloud.com	HTTPS/HTTP	Bind User Domain Name
Static website hosting domain name		HTTPS/HTTP	Configure

7. Enter your Huawei Cloud AK and SK when prompted.

8. Leave the token field blank if you are using permanent AK and SK credentials.

9. Confirm the connection by listing your buckets.

```
& $obsutil ls -s
```



```
Windows PowerShell
Please input your token:
Config file url:
  C:\Users\PC\.obsutilconfig
Update config file successfully!
(base) PS C:\Users\PC>
>> & $obsutil ls -s
Start at 2026-08-17 22:38:17.236657 +0000 UTC

obs://abdulazeez-test
obs://abioye-k8s-lab-obs
obs://bucket-test-ad
obs://dump-cc5
obs://image-buket
obs://mongo-drs
obs://my-obs-bucket-af-south-1
obs://paystack-test
obs://payyyssstack
obs://pfs-azeez
obs://script-bucket-evs
obs://sterling-openshift-hcs-poc
obs://testpaystack
obs://ubuntu-uefi-images
obs://woa-11
obs://wuzhanbin
obs://wuzhanbin11
obs://wwwwees
obs://www2221r
Bucket number: 19
```

If this worked, you should see: Your bucket, for example `obs://dump-cc5`, appears in the list.

WARNING: Never include a screenshot that shows your AK or SK on screen. Redact these values heavily, or leave that part of the screen out entirely.

STAGE 11**Upload the Dump to OBS**

This stage moves your exported dump folder from your Windows workstation up into the OBS bucket.

1. Run the following command in PowerShell to upload the entire dump folder.

```
& $obsutil cp `
"C:\Users\PC\Downloads\app.dev_dump" `
obs://dump-cc5/app.dev_dump/ `
-r `
-f `
-j=20
```

Each option in this command has a specific job.

- -r uploads the entire folder and everything inside it, not just one file.
- -f skips the confirmation prompt for every single file, which saves time on large folders.
- -j=20 uploads up to twenty files at the same time, to speed the transfer up.

2. Wait for the upload to complete.

```
Task id: f9e2f6e3-1c39-4d65-9b02-c1cd3f2eadd7
OutputDir: C:\Users\PC\.obsutil_output

[-----] 100.00% tps:4.82 25.51KB/s 11/11 58.21KB/58.21KB 2.483s
Succeed count: 11 Failed count: 0
Succeed bytes: 58.21KB
Metrics [max cost:2480 ms, min cost:279 ms, average cost:999.45 ms, average tps:4.41, transferred size:58.21KB]
```

If this worked, you should see: The summary reports a Failed count of 0.

3. Confirm the files landed correctly with the following command.

```
& $obsutil ls obs://dump-cc5/app.dev_dump/
```

```
Task id: f9e2f6e3-1c39-4d65-9b02-c1cd3f2eadd7
(base) PS C:\Users\PC> & $obsutil ls obs://dump-cc5/app.dev_dump/
Start at 2026-08-17 22:41:00.5540237 +0000 UTC

Listing objects .

Folder list:
obs://dump-cc5/app.dev_dump/
obs://dump-cc5/app.dev_dump/app.dev_dump/

Object list:
key                                LastModified                    Size    StorageClass    ETag
-----
obs://dump-cc5/app.dev_dump/app.dev_dump/@.done.json 2026-08-17T22:40:45Z          293B    standard        "ef01a037c
15d4ddc900b1826cfbb2881"
obs://dump-cc5/app.dev_dump/app.dev_dump/@.json      2026-08-17T22:40:45Z          54.70KB standard        "3219c2f79
3087756ebc324e153a308e7"
```

STAGE 12

Create a Migration Host on Huawei ECS

ECS, which means: Elastic Cloud Server, is a virtual machine. This stage creates a small Linux server that will act as the bridge between OBS and your RDS database.

1. Open the ECS service in the Huawei Console.
2. Start creating a new ECS instance.
3. Set the region to AF-Johannesburg.
4. Choose a small flavor with 2 vCPUs and 4 GiB of memory.

The screenshot shows the 'Buy ECS' page in the Huawei Cloud Console. The 'Basic Configuration' section is active, showing the following settings:

- Billing Mode:** Pay-per-use (selected)
- Region:** AF-Johannesburg
- AZ Type:** HomeZones
- AZ:** LOS1-AZ1

The 'Configuration Summary' on the right provides a overview of the configuration:

- Basic Configuration:** Billing Mode: Pay-per-use; Region/AZ: AF-Johannesburg | LOS1-AZ1
- Instance:** Specifications: General computing-plus | c7n.large.2 | 2 vCPUs | 4 GiB
- OS:** Image: Mandatory
- Storage:** System Disk: General Purpose SSD, 40 GiB
- Network:** VPC: Mandatory; Primary NIC: Mandatory
- Security Group:** (empty)

5. Select Ubuntu 24.04 64-bit as the operating system image.

The screenshot shows the 'Buy ECS' page with the 'Image' selection step. The 'Public image' tab is selected, and the 'Ubuntu' image is highlighted in the grid. The 'Configuration Summary' on the right shows the OS set to 'Ubuntu 24.04 server 64bit'.

The 'Image' section shows the following options:

- Public image (selected)
- Private image
- Shared image
- KooGallery image

The grid of images includes:

- Huawei Cloud EulerOS
- CentOS
- SUSE
- Ubuntu (highlighted)
- Debian
- OpenSUSE
- Fedora
- AlmaLinux
- Rocky Linux
- CentOS Stream
- CoreOS
- openEuler
- SUSE SAP
- Other

The 'Configuration Summary' on the right shows the following details:

- Basic Configuration:** Billing Mode: Pay-per-use; Region/AZ: AF-Johannesburg | LOS1-AZ1
- Instance:** Specifications: General computing | s7n.large.2 | 2 vCPUs | 4 GiB
- OS:** Image: Ubuntu 24.04 server 64bit
- Storage:** System Disk: General Purpose SSD, 40 GiB
- Network:** VPC: Mandatory; Primary NIC: Mandatory
- Security Group:** (empty)

- 6. Place this ECS in the same VPC as your RDS instance, for example vpc-hw.
- 7. Place this ECS in the same subnet as your RDS instance, for example subnet-homezone.
- 8. Allow inbound SSH access on TCP port 22, limited to your own administrator IP address.

SummaryInbound RulesOutbound RulesAssociated InstancesTag

Some security group rules may not be applied for cloud servers with certain specifications. [Learn more](#)

If the source is set to 0.0.0.0/0 or ::/0, then all external IP addresses are either allowed or denied to access your instances, depending on if the action is Allow or Deny. If the access is allowed, exposing **high-risk ports**, such as port 22, 3389, or 8848, to the public network will leave your instances vulnerable to network intrusions, service interruptions, data leakage, or ransomware attacks. You should only configure known IP addresses for the security group rule.

Add RuleFast-Add RuleDeleteAllow Common PortsBatch Select

[View Security Group Configuration Examples](#)

Select a property or enter a keyword.

Priority	Action	Type	Protocol & Port	Source	Description	Last Modified	Operation
☐ 1	Allow	IPv4	TCP: 3306	192.168.1.53/32	--	Aug 17, 2026 23:53:18 G...	Modify Replicate Delete
☐ 1	Allow	IPv4	TCP: 22	102.89.68.182/32	MY IP	Aug 17, 2026 23:52:59 G...	Modify Replicate Delete
☐ 1	Allow	IPv6	All	rds-migrate	Allows the instances in the security...	Aug 15, 2026 20:17:20 G...	Modify Replicate Delete
☐ 1	Allow	IPv4	All	rds-migrate	Allows the instances in the security...	Aug 15, 2026 20:17:20 G...	Modify Replicate Delete

- 9. Confirm and create the ECS instance.

If this worked, you should see: A new ECS instance listed with a status that changes to Running after a short wait.

STAGE 13**Confirm the ECS Can Reach the RDS Database**

Before moving any data, this stage confirms that your new ECS can actually talk to your RDS database over the private network.

1. Connect to your ECS instance using SSH.
2. Run the following command, replacing the address with your own RDS private IP address.

```
nc -vz 192.168.1.53 3306
```

If this worked, you should see: The terminal prints a line ending in "succeeded!"

WARNING: If this command fails or times out, stop here. Fix the security group and network settings from Stage 6 before continuing, since the restore in later stages will not work without this connection.

STAGE 14**Download the Dump from OBS onto the ECS**

This stage installs obsutil on the Linux ECS, and uses it to pull the dump folder down from OBS.

```

Ubuntu 24.04.4 LTS ecs-dump tty1
ecs-dump login: root
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-106-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Aug 18 06:59:15 AM CST 2026

System load:  0.15           Processes:            129
Usage of /:   8.7% of 39.26GB Users logged in:       0
Memory usage: 7%           IPv4 address for eth0: 192.168.1.238
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Welcome to Huawei Cloud Service

root@ecs-dump:~# nc -vz 192.168.1.53 3306
Connection to 192.168.1.53 3306 port [tcp/mysql] succeeded!
root@ecs-dump:~# _

```

1. On the ECS, run the following commands to download and extract obsutil for Linux.

```

cd /root

wget \
https://obs-community-intl.obs.ap-southeast-
1.myhuaweicloud.com/obsutil/current/obsutil_linux_amd64.tar.gz

tar -xzvf obsutil_linux_amd64.tar.gz

```

2. Enter the extracted folder and make the tool executable.

```

cd obsutil_linux_amd64_*

chmod 755 obsutil

```

3. Configure obsutil the same way as on Windows.

```
./obsutil config -interactive
```

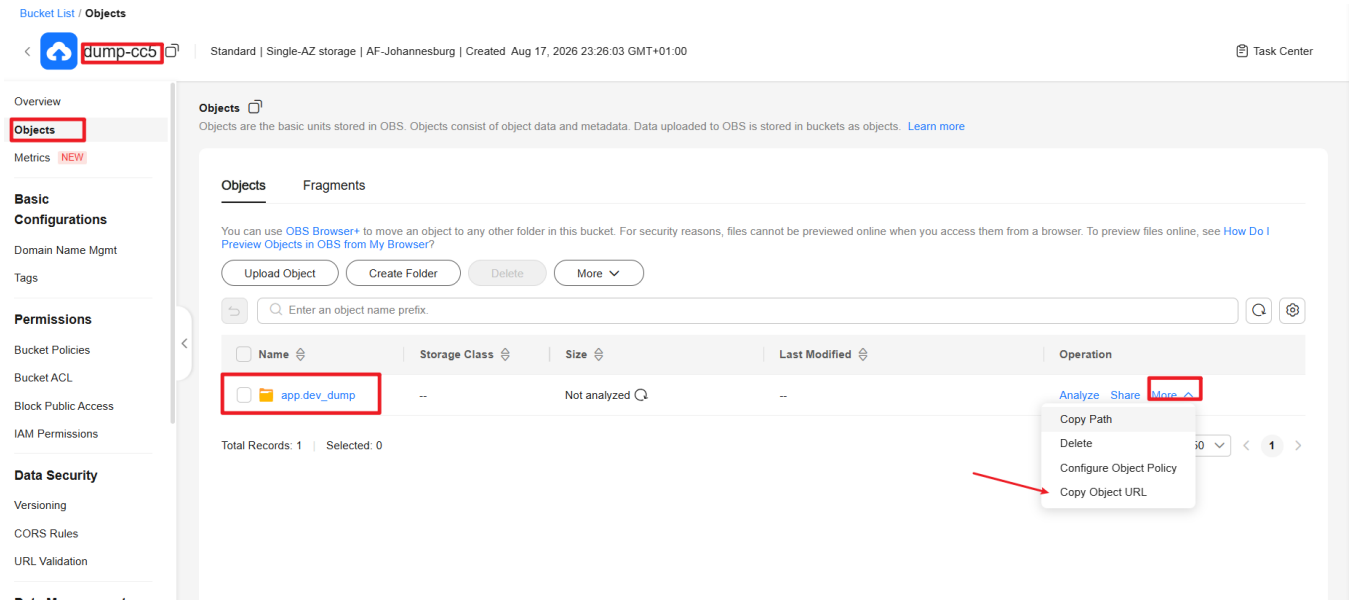
4. Enter the same endpoint used earlier.

```
obs.af-south-1.myhuaweicloud.com
```

5. Create a folder to hold the downloaded dump.

```
mkdir -p /root/app.dev_dump
```

6. Copy the path and Download the dump folder from OBS.



```
./obsutil cp \
  obs://dump-cc5/app.dev_dump/app.dev_dump/ \
  /root/app.dev_dump/ \
  -r \
  -f
```

7. Confirm the files arrived by searching for the dump's metadata file.

```
Find /root -name '@.json' -print
```

If this worked, you should see: The find command prints a path such as /root/app.dev_dump/app.dev_dump/@.json.

NOTE: Because of the extra folder layer mentioned in Stage 11, the real dump files usually end up at /root/app.dev_dump/app.dev_dump. Keep this exact path noted for the next stages.

STAGE 15**Install a Matching MySQL Shell on the ECS**

Just as in Stage 3, the tool used to load the dump must match the MySQL version that created it. This stage installs MySQL Shell 26.7 on the Linux ECS.

1. On the ECS, run the following command to download MySQL Shell 26.7.

```
cd /root

wget \
https://cdn.mysql.com/Downloads/MySQL-Shell/mysql-shell-26.7.0-linux-glibc2.28-x86-64bit.tar.gz
```

2. Extract the downloaded file.

```
Tar -xzf \
mysql-shell-26.7.0-linux-glibc2.28-x86-64bit.tar.gz
```

3. Move the extracted folder into a permanent location.

```
Mv \
mysql-shell-26.7.0-linux-glibc2.28-x86-64bit \
/opt/mysql-shell-26.7
```

4. Create a shortcut link so the mysqlsh command works from anywhere.

```
Ln -sf \
/opt/mysql-shell-26.7/bin/mysqlsh \
/usr/local/bin/mysqlsh
```

5. Confirm the installed version.

```
Mysqlsh -version
```

If this worked, you should see: The command reports MySQL Shell 26.7.0.

STAGE 16**Connect from the ECS to Huawei RDS**

This stage confirms that MySQL Shell on the ECS can log in to your Huawei RDS database, right before the actual restore.

1. Run the following command on the ECS.

```
Mysqlsh \
--sql \
-h 192.168.1.53 \
-P 3306 \
-u root \
-p
```

2. Enter your RDS administrator password when prompted.
3. Run the following command to confirm the destination version.

```
SELECT VERSION();
```

```
root@ecs-dump:~# mysqlsh --sql -h 192.168.1.53 -P 3306 -u root -p
Please provide the password for 'root@192.168.1.53:3306': *****
Save password for 'root@192.168.1.53:3306'? [Y]es/[N]o/[e]ver (default No): Y
MySQL Shell 26.7.0

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Type '\help' or '\?' for help; '\quit' to exit.
Creating a session to 'root@192.168.1.53:3306'
Fetching global names for auto-completion... Press ^C to stop.
Your MySQL connection id is 4314
Server version: 8.0.43-260600 MySQL Community Server - (GPL)
No default schema selected; type \use <schema> to set one
WARNING: This account uses a deprecated authentication method. Please read \help mysql_native_password help
nation.
MySQL 192.168.1.53:3306 SQL > SELECT VERSION();
+-----+
| VERSION() |
+-----+
| 8.0.43-260600 |
+-----+
1 row in set (0.0003 sec)
MySQL 192.168.1.53:3306 SQL > \js
Switching to JavaScript mode...
MySQL 192.168.1.53:3306 JS >
MySQL 192.168.1.53:3306 JS > util.loadDump(
->   "/root/app.dev_dump/app.dev_dump",
->   {
->     threads: 4
->   }
-> )_
```

If this worked, you should see: The version command returns 8.0.43-260600.

STAGE 17**Restore the Dump into Huawei RDS**

This is the stage where your data actually moves into Huawei RDS. Because the source and destination are different MySQL versions, expect one blocked attempt before the real restore.

1. While connected to RDS with MySQL Shell, switch to JavaScript mode.

```
\js
```

2. Attempt the first restore with the following command.

```
util.loadDump(
  "/root/app.dev_dump/app.dev_dump",
  {
    threads: 4
  }
)
```

3. Read the error message that appears.

```
Loading DDL and Data from '/root/app.dev_dump/app.dev_dump' using 4 threads.
Opening dump - done
Dump is complete.
Target is MySQL 8.0.43-260600. Dump was produced from MySQL 26.7.0-cloud
ERROR: Destination MySQL version is older than the one where the dump was created. Loading dumps from non-co
L versions is not fully supported and may not work. Enable the 'ignoreVersion' option to load anyway.
MySQL version mismatch (MYSQLSH 53011)
MySQL 192.168.1.53:3306 JS > _
```

This is what happens if you do not put ignore version to be **True**.

Destination MySQL version is older than the one where the dump was created.

Enable the 'ignoreVersion' option to load anyway.

NOTE: This error is expected in this lab. MySQL Shell is protecting you by default, because loading data from a newer MySQL version into an older one can sometimes cause problems.

4. Run the restore again, this time adding the ignoreVersion option.

```
util.loadDump(
  "/root/app.dev_dump/app.dev_dump",
  {
    threads: 4,
    ignoreVersion: true
  }
)
```

```
MySQL 192.168.1.53:3306 JS > util.loadDump(  
  ->   "/root/app.dev_dump/app.dev_dump",  
  ->   {  
  ->     threads: 4,  
  ->     ignoreVersion: true  
  ->   }  
  -> )
```

5. Wait for the restore to finish.

If this worked, you should see: The restore completes without further errors, and MySQL Shell reports the tables and rows that were loaded.

WARNING: The ignoreVersion option does not convert or fix features that are incompatible between MySQL versions. It only tells MySQL Shell to attempt the load anyway. Only use this option after you have checked that your database does not depend on features unique to the newer MySQL version.

STAGE 18**Validate the Finished Migration**

This final stage proves, step by step, that the data really did arrive safely in Huawei RDS.

1. Open Huawei DAS and log in to your RDS instance.
2. Run the following command to confirm the database exists.

```
SHOW DATABASES LIKE 'app.dev';
```

3. Switch into that database.

```
USE `app.dev`;
```

4. List the tables inside it.

```
SHOW TABLES;
```

5. Read the data inside the users table.

```
SELECT * FROM users;
```

The screenshot shows the Huawei DAS SQL Window interface. The 'Current Database' is set to 'app.dev'. The SQL editor contains the following commands:

```
1 SHOW DATABASES LIKE 'app.dev';
2
3 USE `app.dev`;
4
5 SHOW TABLES;
6
7 SELECT * FROM users;
```

The execution results for the 'SELECT * FROM users;' command are displayed in a table with the following data:

	id	name	email	created_at
1	1	Toba	toba@example.com	2026-08-15 19:42:33
2	2	Test User	test@example.com	2026-08-15 19:42:33

If this worked, you should see: The users table contains the same rows that existed on the OCI source, including their original timestamps.

SUCCESS: If you see both original rows with their original timestamps, the migration is complete and verified end to end.

What to do if something goes wrong

1. MySQL Shell refuses to export the database

If you see "Unsupported MySQL Server" during export, your MySQL Shell tool is older than your source database. Install a MySQL Shell version that matches your source, as shown in Stage 3, then try the export again.

2. The ECS cannot reach the RDS database

If the "nc -vz" command from Stage 13 times out or fails, check that both the ECS and the RDS instance are in the same VPC and subnet. Then check that the RDS security group allows inbound TCP port 3306 from the ECS.

3. util.loadDump() is blocked by a version mismatch

If you see "Destination MySQL version is older than the one where the dump was created," this is expected when the source is newer than the destination. Review your schema for features unique to the newer version, then retry with ignoreVersion: true as shown in Stage 17.

4. obsutil cannot list your buckets

If "obsutil ls -s" returns an authentication error, re-run "obsutil config -interactive" and re-enter your AK and SK carefully. Confirm you are using the correct regional endpoint, for example obs.af-south-1.myhuaweicloud.com.

5. The database name app.dev seems to cause problems

If a migration tool rejects the name app.dev, this is very likely a limitation of that specific tool, not of MySQL itself. Stage 8 in this guide confirms that Huawei RDS for MySQL accepts this name when it is written with backticks, for example `app.dev`.

WARNING: Once your migration is validated, remove temporary resources: delete the dump folder from the ECS, delete the dump objects from OBS according to your retention policy, remove temporary security group rules, and terminate the migration ECS if it is no longer needed.

Quick recap

By completing this lab, you accomplished the following.

- Opened a secure Bastion tunnel into a private OCI MySQL database.
- Exported the database into a logical dump folder using MySQL Shell.
- Created a Huawei RDS for MySQL destination in the AF-Johannesburg region.
- Confirmed that Huawei RDS accepts a database name containing a period.
- Moved the dump folder through Huawei OBS using obsutil.
- Created a Huawei ECS migration host on the same private network as the RDS instance.
- Restored the dump into Huawei RDS using MySQL Shell, working around a version mismatch with `ignoreVersion`.
- Validated that all original data and timestamps arrived safely.

Things to remember:

- Always match your MySQL Shell version to your source database's version.
- Never expose a production database port to the whole internet; scope security group rules tightly.
- The `ignoreVersion` option skips a warning, but it does not fix version-specific compatibility problems for you.
- Redact AK/SK values and passwords from every screenshot and document.
- Clean up temporary migration resources once the migration is verified.